Consistency-Corrected Demosaicing: Recovering High-Frequency Detail from Bilinear Reconstruction*

Tim Meekins
tmeekins42@gmail.com

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Abstract

Bilinear interpolation of a Bayer color filter array is a correct low-pass reconstruction: its error largely vanishes under downsampling. What it discards is high-frequency detail — but that detail is not lost, since every sensel is an exact measurement that bilinear has averaged away. We describe a method that recovers it. A leave-one-out residual is computed at each site by predicting that site's own measured channel from the reconstruction at its same-color neighbors, excluding the sample itself; the residual is diffused off the CFA lattice with edge-aware weighting, and applied as a single multiplicative luminance factor common to all three channels. Because the correction is a common scale factor it preserves channel ratios exactly and is therefore structurally incapable of introducing false color, in contrast to additive per-channel corrections. Measured across ten frames from three camera bodies spanning ISO 100–12800, the method recovers more high-frequency detail than adaptive homogeneity-directed demosaicing (median 174% against 150% of bilinear at base ISO) at comparable or lower chroma noise. We report the negative results alongside the positive ones, and describe two measurement failures that produced confidently wrong conclusions before they were caught.

1 Introduction

A Bayer sensor records one color channel per photosite. Demosaicing reconstructs the two missing channels at every site, and the difficulty is concentrated entirely in high-frequency content: in a smooth region every method agrees, and the differences between methods are confined to edges and fine texture.

The observation motivating this work is that bilinear interpolation is not merely a weak method but a *correct* one in a specific and useful sense. Averaging the neighboring samples estimates the local mean well; downsample a bilinear reconstruction and its error largely disappears. Its failure is band-limited — it discards detail above the sampling frequency of each channel.

That detail, however, has been measured. Every sensel holds an exact reading of one channel, and bilinear has smeared it across a neighborhood. This suggests recovering the discarded detail from the samples themselves rather than inventing it by sharpening. The distinction matters: a sharpening kernel amplifies whatever survived the blur, including noise, with no knowledge of the true value. The method described here is driven entirely by the sensor data.

2 Note on authorship and tooling

The method described here — treating bilinear interpolation as a correct low-pass reconstruction, recovering the discarded detail from the sensor's own samples, and applying that correction so as to preserve hue — originated with the author, as did the visual assessments on which the qualitative claims rest.

A large language model was used as a collaborator throughout. Its contribution was the implementation in C++ and HLSL, the design and coding of the measurement tools, the literature search reported in Section 8, and the drafting of this note. Several of the design decisions recorded here were also arrived at in that dialogue, in both directions: the diagnosis that an isotropic correction cannot recover lattice-sampled detail is one; the identification of the negative-channel mechanism in Section 3.4 is another; and

*A reference implementation, together with the measurement tools used to produce every figure and table in this note, is available at https://github.com/tmeekins42/tglab.

the observation that the sampled channel is exact, which bounds what the consistency constraint can yield at all, came from the author.

This is disclosed for two reasons. The first is ordinary honesty about provenance. The second is more specific to the content: Section 7 reports two measurement failures in which a plausible quantitative result contradicted the rendered image, and in both cases the error was introduced by the model and caught by human inspection of the pictures. A reader weighing the evidence here should know which parts were machine-generated and are therefore subject to that failure mode, and which rest on a photographer looking at photographs.

3 Method

3.1 The consistency constraint

Any reconstruction can be re-sampled through the CFA and compared against the original mosaic. Where the two differ, the reconstruction is provably wrong — no ground truth and no assumption about scene content is required.

Applied naively this measures nothing. Every practical demosaicing method copies the sampled channel verbatim, so consistency at the sampled sites is satisfied by construction. The useful form is *leave-one-out*: predict a site's own channel from its neighbors while withholding the sample there. That prediction can be wrong, and how wrong it is quantifies exactly what bilinear discarded.

Let $S(x, y)$ denote the normalized mosaic sample and c the CFA color at (x, y). Let P_c be the bilinear reconstruction of channel c. The residual is

$$\Delta(x,y) \;=\; S(x,y) \;-\; \tfrac{1}{4}\big[\,P_c(x{-}2,y) + P_c(x{+}2,y) + P_c(x,y{-}2) + P_c(x,y{+}2)\,\big]. \tag{1}$$

The offsets are ± 2 because the Bayer period is two: those four neighbors are the nearest sites of the same color.

3.2 Relation to Malvar–He–Cutler

Equation 1 is not new. It is, term for term, the correction used by Malvar, He and Cutler [1], who describe the same quantity as an estimated luminance change and add a gain-controlled portion of it to the interpolated value. We arrived at it independently from the consistency argument above, which is some evidence that the reasoning is sound, but it is not a contribution.

The contribution, such as it is, lies in what follows: how the residual is distributed, and how it is applied.

3.3 Edge-aware diffusion

The residual of Equation 1 is defined only at each site's own color, and is therefore known on the CFA lattice — at half of sites for green and a quarter each for red and blue. Applying it directly would correct in a phase-correlated pattern and reintroduce precisely the lattice artifact that adaptive methods exist to remove.

The residual is therefore diffused across a 3×3 neighborhood, weighted so that samples on the far side of an edge are excluded:

$$\tilde{\Delta}(x,y) \;=\; \frac{\sum_{j\in N} w_j\,\Delta_j}{\sum_{j\in N} w_j}, \qquad w_j = \begin{cases} 1 & j=(x,y) \\ \frac{1}{2} & |L_j - L_{(x,y)}| \le \tau \\ 0 & \text{otherwise,} \end{cases} \tag{2}$$

where L is bilinear's luminance and $\tau = \tau_0 + \tau_1 L_{(x,y)}$. The tolerance is proportional to the local level plus a floor, so that the same test means the same thing in shadow and highlight; an absolute threshold silently stops discriminating over most of the tonal range. Bilinear's luminance is a suitable guide here because its *edges* are correctly located — it is the detail that is missing, not the structure.

3.4 Multiplicative luminance injection

The corrected pixel is obtained by scaling all three channels by a single factor:

$$\mathbf{c}' \;=\; k\,\mathbf{c}, \qquad k \;=\; \mathrm{clamp}\!\left(\frac{L + s\,\tilde{\Delta}}{L},\; k_{\min},\; k_{\max}\right), \tag{3}$$

where s is a user strength and L the reconstructed luminance.

Two properties follow. First, since k multiplies every channel equally, the ratios R/G and B/G are preserved exactly. This is constant-hue in the sense of Freeman [2], and it means the correction *cannot* introduce false color: no scaling of a triple can change which channel dominates. An additive per-channel correction, such as Malvar–He–Cutler's, has no such guarantee.

We note for precision that scaling preserves *ratios*, not *differences*: k maps $R - G$ to $k(R - G)$. Preserving differences exactly would require a common additive offset instead. Ratio preservation is the appropriate invariant, since an additive offset shifts hue in shadow where one channel approaches zero and a fixed offset is a large relative change.

Second, the clamp on k bounds the correction. An unbounded injection rings at hard edges, which is the characteristic unpleasantness of sharpening and precisely what this method exists to avoid.

Finally, the reconstruction is bounded by the range of real samples of the same color in the 3×3 neighborhood. This is necessary rather than defensive: camera color matrices have large negative off-diagonal terms — a Canon EOS R5's first row is $[1.535, -0.555, 0.019]$ — so a channel that is merely *short* relative to green becomes negative after the matrix, clamps at zero, and renders fully saturated.

3.5 Chroma median

A 3×3 median is applied to the color differences $R - G$ and $B - G$, gated on luminance similarity by the same rule as the diffusion step. Filtering the differences rather than the channels leaves luminance detail — the object of the entire method — untouched.

Median filtering of color differences after demosaicing is long-established practice, introduced by Freeman [2] and exposed by dcraw as its `-m` option. Its inclusion here is not a contribution; its absence, however, was the one axis on which the method initially underperformed (Section 5).

3.6 Reduction to bilinear

At $s = 0$ with the median disabled, Equation 3 gives $k = 1$ and the method reduces exactly to bilinear. This is enforced by a regression test and is what makes the measurements below trustworthy: any divergence at zero strength is a defect in the correction rather than a property of it.

4 Experimental method

4.1 Why a single error figure is inadequate

The natural evaluation — mean squared error against a reference — is actively misleading for this problem, and we report two measurements instead.

Squared error rewards not being wrong; it does not reward being sharp. A smooth reconstruction scores well by declining to commit, and on fur, feathers and foliage a slightly misplaced strand reads better to a human observer than a correctly averaged smudge. This is not a rationalisation after an unfavorable result: it is visible in the data, since the method described here scores worse than AHD on squared error against a downsampled reference while producing demonstrably more high-frequency energy.

A second difficulty is that any reference derived by downsampling a demosaiced frame has already destroyed sub-pixel content — a specular glint on a dew drop, occupying one or two sensels — which a method that preserves such content is then penalised for reproducing.

We therefore measure two quantities separately:

Detail the median Laplacian magnitude, normalized by local level, in *textured* regions (local range $\geq 60\%$ of the local mean).

Noise the median local standard deviation, luminance and chroma separately, in *flat* regions (local range $\leq 25\%$ of the local mean), where any high-frequency content is noise by definition.

Both thresholds are relative to the local level so that they mean the same thing across the tonal range. All figures are reported as a percentage of bilinear on the same frame, which removes scene content as a variable.

4.2 Data

Ten raw frames from three camera bodies (Canon EOS R5, Canon EOS 5D Mark III, Sony), spanning ISO 100 to ISO 12800, covering portraiture, foliage, architecture and backlit fine structure.

5 Results

5.1 Quantitative

Table 1 reports detail and noise for all four methods, each relative to bilinear on the same frame.

Table 1: Detail and noise relative to bilinear (= 100) on the same frame. Detail: higher is better. Noise: lower is better. The best entry in each column group is set in bold; bilinear is excluded from that comparison, since it is 100 everywhere by construction as the baseline the other figures are expressed against. Ties are bolded jointly.

		Bilinear			Malvar			AHD			Proposed		
Frame	ISO	Det	Lum	Chr	Det	Lum	Chr	Det	Lum	Chr	Det	Lum	Chr
_U0A1067.CR3	4000	100	100	100	148	102	89	147	**101**	77	**157**	102	**73**
_U0A3431.CR3	12800	100	100	100	145	102	102	144	**100**	90	**148**	102	**74**
_U0A3636.CR3	3200	100	100	100	**173**	**102**	88	163	**102**	76	163	109	**70**
_U0A5083.CR3	100	100	100	100	150	104	87	157	**102**	74	**191**	111	**72**
_U0A6675.CR3	3200	100	100	100	158	103	89	157	**102**	77	**162**	106	**72**
_U0A6757.CR3	4000	100	100	100	**172**	102	88	167	**101**	76	170	108	**66**
_U0A6810.CR3	100	100	100	100	145	103	88	150	**102**	74	**174**	106	**72**
_MG_9673.CR2	1000	100	100	100	**178**	127	89	172	**125**	76	169	152	**71**
_DSC0017.ARW	125	100	100	100	168	105	90	**175**	**103**	79	**175**	117	**75**
_DSC0162.ARW	125	100	100	100	184	106	89	204	**105**	77	**234**	122	**74**

The proposed method recovers the most detail on seven of ten frames and has the lowest chroma noise on all ten. It has the lowest luminance noise on none: that figure is consistently higher than AHD's, at 102–122% of bilinear against AHD's 100–105% across nine of the ten frames, with the tenth (`_MG_9673`, 152% against 125%) discussed below. This is the expected consequence of restoring isolated bright samples: the method cannot distinguish a genuine sub-pixel glint from a noise spike, and preserves both.

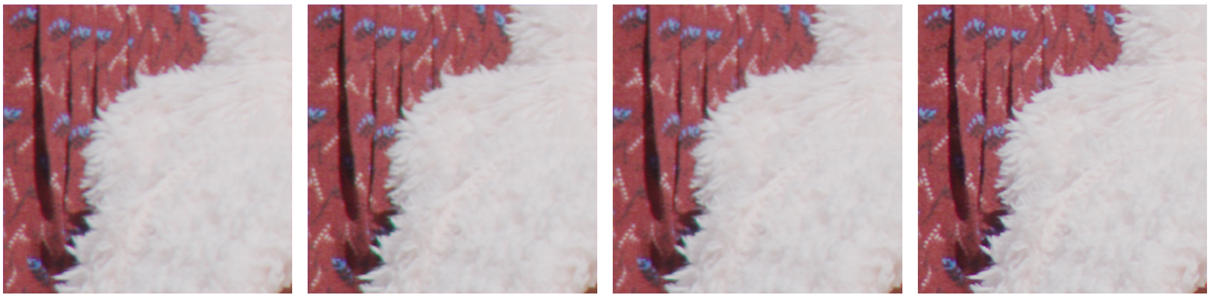
Two observations from the table deserve comment. The detail advantage narrows as ISO rises (174% against AHD's 150% at ISO 100; 148% against 144% at ISO 12800), which follows from the same mechanism: at high ISO a growing fraction of the isolated samples being restored are noise rather than detail. And frame `_MG_9673` at ISO 1000 is an outlier in luminance noise (152%), the only frame where the method's noise cost is severe; we have not determined why.

5.2 Qualitative

Figures 1, 2 and 3 show crops from a single base-ISO portrait (300 × 300 and 320 × 320 pixels at native resolution), developed identically through a filmic tone curve and reproduced without resampling.

5.3 Cost

At 45 megapixels on a consumer GPU the method completes in 598 ms, against 567 ms for AHD and 545 ms for Malvar. All three are bandwidth-bound rather than compute-bound at this size, so the additional passes are nearly free.



(a) Bilinear (b) Malvar (c) AHD (d) Proposed

Figure 1: A white fleece collar against a printed red garment: a high-contrast, high-chroma boundary crossed by fur strands finer than the sampling interval. Bilinear shows a cyan-green fringe along the boundary and mottling within the printed fabric; the proposed method resolves individual strands and leaves the boundary neutral. This is the case that motivated the work.

(a) Bilinear (b) Malvar (c) AHD (d) Proposed

Figure 2: Individual hair strands against a defocused background — structures at or below the sampling limit of the green channel. Bilinear shows pronounced cyan and magenta speckling through the hair mass.

(a) Bilinear (b) Malvar (c) AHD (d) Proposed

Figure 3: Printed fabric: a fine repeating pattern in saturated blue on red, near the resolution limit. High-frequency chromatic content of this kind is where demosaicing methods characteristically produce false color.

6 Negative results

We report the following because they are the part of this work least likely to exist elsewhere, and because each cost material effort.

6.1 Consistency steering does not work

Since the residual measures which reconstructions are wrong, it is natural to use it to *choose* an interpolation direction rather than to correct one afterwards — replacing AHD's least-gradient criterion with a best-prediction criterion. This was measured before implementation, over 2.4 million decisive sites on two frames: the two criteria disagree on approximately 45% of sites, and where they disagree the gradient criterion is correct 61–72% of the time.

The specific case motivating the idea also failed. On textured sites, where horizontal and vertical gradients are nearly equal and the gradient criterion should have least information, gradient still won 56–63%. The explanation is that smoothness of $C - G$ is not equivalent to correctness of G: a systematically wrong green produces a perfectly smooth difference plane, because the error cancels between neighbors.

6.2 The correction saturates

Applying more than approximately half the measured residual degrades squared error, and full strength returns roughly to bilinear's error level. The cause is the diffusion step: because the residual is known only on the CFA lattice, spreading it is a blur, and that blur discards much of the high frequency the correction exists to restore. Beyond half strength, more of a blurred correction adds error faster than it adds detail.

The deeper limitation is that the sampled channel is exact — every method copies it verbatim — so consistency at a site is free and carries no information. The only available leverage is indirect, through channels never measured at that site, and indirect evidence proves thin.

7 Methodological failures

Two measurement errors produced confident and wrong conclusions during this work. Both are easy to repeat.

Aggregate statistics disagreed with the rendered image, in both directions. Removing the chroma median's edge gate improved the mean magenta index in the dark deciles from $+0.031$ to -0.005, better than the reference method — while visibly painting green streaks along every fine strand. A per-decile mean averages over the whole frame and cannot see a defect that is thin, bright and localised, which was the only defect that mattered. Scoring at edge sites, as a high percentile, and inspecting a rendered crop are all necessary.

A fixture that cannot express a failure cannot test for it. A synthetic test scene used an identity color matrix, under which the negative-channel mechanism of Section 3.4 *cannot occur*. The corresponding test passed identically with the fix present and with it removed. Separately, a regression test for reduction to bilinear used a linear-gradient scene, which bilinear reconstructs exactly; the residual was therefore zero and the test could not fail regardless of how badly the strength parameter was mishandled. Both were caught only by deliberately breaking the implementation and confirming the test noticed.

8 Relation to prior work

The residual of Equation 1 is that of Malvar–He–Cutler [1]. The chroma median postprocessing step is Freeman's [2]. Neither is claimed here.

Residual Interpolation [3, 4] shares terminology but differs mechanically: its tentative estimate comes from a guided filter regressing one channel on green, so its residual is a cross-channel model error rather than a same-channel sampling error; it is computed without leave-one-out exclusion, interpolated linearly rather than edge-aware, and added per-channel rather than injected multiplicatively.

Alternating-projection methods [5] enforce CFA consistency, but by hard reinsertion of the known samples — an operation that constrains the sampled sites and does nothing to their neighbors. The diffusion step here is what those methods lack in a single pass.

The multiplicative injection of Equation 3 is structurally the Brovey-style detail injection used in pansharpening, where preserving band ratios by construction is standard; the connection between CFA demosaicing and pansharpening has been made explicitly [7]. Its nearest relative within demosaicing is the local color ratio postprocessing of Lukac *et al.* [6], which also preserves ratios against the original CFA data but corrects channels sequentially through green rather than by a single common gain.

We were unable to find this particular composition — a Malvar-style residual, diffused with edge awareness, injected as a common multiplicative luminance factor — in the literature. The honest characterisation is a novel combination of established components rather than a new principle. The one structural claim we would defend is that of Section 3.4: the correction is incapable of introducing false color, where an additive per-channel correction is not.

We note also that US 7,502,505 B2 (Microsoft, filed 2004) claims computing the gradient of Equation 1 and adding a gain-controlled portion to an interpolated value. On a non-specialist reading its claims do not appear to cover edge-aware diffusion or multiplicative all-channel scaling, but this should not be relied upon.

9 Conclusion

Treating bilinear interpolation as a correct low-pass reconstruction, and recovering the high-frequency content it discards from the sensor's own measurements, produces a demosaicing method competitive with adaptive alternatives at comparable cost. Its principal advantage is structural: because the correction is a common multiplicative factor it cannot introduce false color. Its principal cost is that it cannot distinguish a genuine sub-pixel highlight from a noise spike, and consequently preserves both — an increasingly unfavorable trade as sensitivity rises.

A Implementation

The complete method, in the order the reference implementation performs it. All quantities are in normalized sensor units: the mosaic sample S is $(\text{raw} - \text{black})/(\text{white} - \text{black})$, clamped to $[0, 4]$ — the upper bound exceeds unity because white balance is applied after reconstruction and legitimately pushes channels above the white level.

$\text{CFA}(x, y) \in \{0, 1, 2\}$ gives the color sampled at a site, and $L(\mathbf{c}) = 0.2126r + 0.7152g + 0.0722b$.

Algorithm 1 Consistency-corrected demosaicing.

Require: mosaic S, strength s, median passes m
Ensure: reconstructed image $\mathbf{C}$

Stage 1 — bilinear reconstruction
for each site (x, y) **do**
  $c \leftarrow \text{CFA}(x, y)$; $\mathbf{C}[c] \leftarrow S(x, y)$
  if $c = 1$ **then** ▷ green site: the other two lie on opposite axes
    $h \leftarrow \text{CFA}(x-1, y)$
    $\mathbf{C}[h] \leftarrow \frac{1}{2}\big(S(x-1, y) + S(x+1, y)\big)$
    $\mathbf{C}[2-h] \leftarrow \frac{1}{2}\big(S(x, y-1) + S(x, y+1)\big)$
  else ▷ red or blue site
    $\mathbf{C}[1] \leftarrow \frac{1}{4}\big(S(x\pm1, y) + S(x, y\pm1)\big)$
    $\mathbf{C}[2-c] \leftarrow \frac{1}{4}\big(S(x\pm1, y\pm1)\big)$ ▷ four diagonal neighbors
  end if
end for

Stage 2 — leave-one-out residual
for each site (x, y) **do**
  $c \leftarrow \text{CFA}(x, y)$
  $p \leftarrow \frac{1}{4}\big(\mathbf{C}_c(x\pm2, y) + \mathbf{C}_c(x, y\pm2)\big)$ ▷ this site's own sample excluded

$\quad\Delta(x,y) \leftarrow S(x,y) - p$
end for

Stage 3 — edge-aware diffusion
for each site (x,y) **do**
$\quad\ell \leftarrow L(\mathbf{C}(x,y)); \quad \tau \leftarrow \tau_0 + \tau_1 \ell$
$\quad a \leftarrow 0; \quad W \leftarrow 0$
$\quad$**for** each j in the 3×3 neighborhood **do**
$\quad\quad$**if** $|L(\mathbf{C}_j) - \ell| \le \tau$ **then**
$\quad\quad\quad w \leftarrow 1$ if $j = (x,y)$ else $\frac{1}{2}$
$\quad\quad\quad a \leftarrow a + w\,\Delta_j; \quad W \leftarrow W + w$
$\quad\quad$**end if**
$\quad$**end for**
$\quad$**if** $W > 0$ **then**
$\quad\quad\tilde{\Delta}(x,y) \leftarrow a/W$
$\quad$**else**
$\quad\quad\tilde{\Delta}(x,y) \leftarrow 0$
$\quad$**end if**
end for

Stage 4 — injection and bounding
for each site (x,y) **do**
$\quad\ell \leftarrow L(\mathbf{C}(x,y))$
$\quad$**if** $\ell > \epsilon$ **then**
$\quad\quad k \leftarrow \mathrm{clamp}\big((\ell + s\tilde{\Delta}(x,y))/\ell,\ k_{\min},\ k_{\max}\big)$
$\quad\quad\mathbf{C}(x,y) \leftarrow k\,\mathbf{C}(x,y)$ ▷ common factor: ratios preserved
$\quad$**end if**
$\quad$**for** $c \in \{0,1,2\}$ **do**
$\quad\quad[\,lo_c, hi_c\,] \leftarrow$ range of S over 3×3 sites where CFA $= c$
$\quad\quad\mathbf{C}_c \leftarrow \mathrm{clamp}(\mathbf{C}_c,\ lo_c,\ hi_c)$ ▷ see §3.4
$\quad$**end for**
end for

Stage 5 — chroma median
$D_r \leftarrow \mathbf{C}_r - \mathbf{C}_g; \quad D_b \leftarrow \mathbf{C}_b - \mathbf{C}_g$
for m passes, and for each of D_r, D_b **do**
$\quad$**for** each site (x,y) **do**
$\quad\quad V \leftarrow \{\, D_j : j \in 3 \times 3,\ |L(\mathbf{C}_j) - \ell| \le \tau \,\}$
$\quad\quad$**if** $|V| \ge 3$ **then**
$\quad\quad\quad D(x,y) \leftarrow \mathrm{median}(V)$
$\quad\quad$**end if** ▷ fewer than three is not a median
$\quad$**end for**
end for
$\mathbf{C}_r \leftarrow \mathbf{C}_g + D_r; \quad \mathbf{C}_b \leftarrow \mathbf{C}_g + D_b$ ▷ green untouched: luminance detail survives

Stage 6 — color
record which channels reached the clipping level, *before* gains
$\mathbf{C} \leftarrow \mathbf{C} \odot \mathrm{camMul}$
restore clipped channels to the maximum of the three
$\mathbf{C} \leftarrow M_{\mathrm{cam}}\,\mathbf{C}; \quad \mathbf{C} \leftarrow \max(\mathbf{C}, 0)$

Constants in the reference implementation: $\tau_0 = 0.004$, $\tau_1 = 0.12$, $k_{\min} = 0.5$, $k_{\max} = 2.0$, $\epsilon = 10^{-5}$, clipping level 0.99. Defaults are $s = 1$ and $m = 1$.

Two details are worth stating because they are easy to get wrong. Stage 4 must precede stage 5: filtering the color differences before the samples have been bounded lets the median select an out-of-range value that bounding would have rejected, and during development this ordering error made the CPU and GPU paths disagree by 0.008 — far above half-float rounding, and caught by a cross-check rather than by eye. And stage 6's clipping decision must be taken before the white-balance gains are applied but repaired after, since "clipped" is a statement about the sensor while "neutral" is a statement about the balanced image.

The GPU implementation performs the same six stages as four compute kernels over shared scratch planes, with the chroma median ping-ponging between two pairs of planes: a pass may not read and write the same buffer, since threads within a dispatch have no ordering.

References

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